CANDIDATES EVALUATION TASK

Task Title: Financial News Analysis and Stock Price Prediction

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Objective

To assess the candidate's ability to:

1. Perform EDA on a financial dataset.
2. Extract stock names or ticker symbols using NLP.
3. Retrieve financial data using APIs or publicly available datasets.
4. Forecast stock prices and analyze the impact of news on stock financials gathered from step 3.

NOTE: The evaluation of the report will be based on your approach and your ability to explain your solution. There can be more than one solutions.

Dataset Description

- **Headlines**: Titles of financial news articles.
- **Date**: Publication date of the news article.
- **Description**: A detailed text description of the news article.

Task Instructions

1. Exploratory Data Analysis (EDA)

- Analyze the dataset for basic properties such as:
    - The number of unique articles.
    - Trends over time (e.g., volume of articles by date).
    - Patterns in headlines or descriptions (e.g., word frequencies).
- Handle any missing or inconsistent data, and justify your preprocessing steps.

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2. Stock Ticker Extraction

- Extract company names or stock tickers (e.g., AAPL for Apple Inc., TSLA for Tesla).
- If the description contains a company name but no stock ticker, map the company name to its stock ticker using a reliable mapping source or API (e.g., Yahoo Finance, Alpha Vantage, publicly available datsets).
- Provide a list of extracted stock tickers with corresponding news articles.

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3. Financial Data Retrieval

- Use Yahoo Finance, Alpha Vantage, or any publicly available dataset to retrieve the following details for each stock extracted from the step above:
    - Historical price data (close prices for the past year).
    - Key financial metrics (e.g., market capitalization, P/E ratio).
- Ensure proper API integration or dataset usage with error handling for missing data.

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4. Stock Price Forecasting

- Predict the stock prices for a specified time period (e.g., the next 7, 15, or 30 days).
- Use any forecasting method of your choice, such as:
    - Time-series models.
    - Machine learning techniques.
- Evaluate the model's performance using appropriate metrics.
- Visualize the forecast alongside historical data.

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5. News Analysis and Impact Evaluation

- Analyze the content of each news article and determine its potential impact on the financial performance of the stock.
- Correlate key aspects of the news (e.g., keywords, topics, tone) with stock price movements or other financial indicators.
- Provide insights on:
    - How the news might influence the stock's financials.
    - The level of confidence in your predictions based on historical trends and patterns.

Deliverables

The candidate should submit the following:

1. A comprehensive report detailing:
    - The EDA findings.
    - The methodology for stock ticker extraction.
    - Financial insights and forecasting results.
    - Correlation of the news with stock performance.
    - Conclusion and recommendations.
2. Annotated Python code, well-documented, and following best practices.
3. Visualizations to support findings (e.g., stock trends, forecast graphs, sentiment distributions).
4. A README file explaining the steps to execute the code.

Timeline

The task must be completed and submitted within **5 days (Wednesday, 22nd January 25, 1500hrs)**.

Additional Notes

- Candidates are encouraged to use open-source tools and libraries.
- Use of creative techniques and clear presentation of insights will be considered a plus.
- Candidates may use any publicly available datasets or APIs.
- Assume and list any missing data, anomaly, ambiguity or confusion.
- Send your submissions as a github repo to danish.jilani19@gmail.com
- In case of confusion, feel free to contact awaisnawaz2000@gmail.com